3 **(a)** Describe and explain the trend in the thermal stability of the hydrogen halides HC*l*, HBr and HI. Include an equation for the thermal decomposition reaction in your answer. **[3]**

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(b) To form concentrated solutions of hydroiodic acid, hydrogen iodide gas is first formed by reaction of hydrogen and iodine gas before being bubbled into water.

$$H_2(g) + I_2(g) \rightleftharpoons 2HI\ (g)$$

(i) State the conditions necessary for a gas to approach ideal behaviour. **[1]**

(ii) The graphs of *pV/RT* against *p* for HI gas and gas **J** are shown in Fig. 3.1.

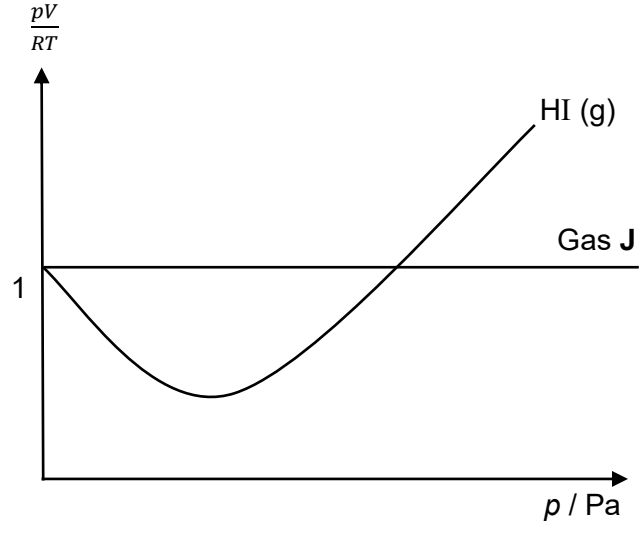
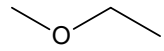


Fig. 3.1

Explain the shape of the graph for HI gas and gas **J**. Your answer should include references to intermolecular forces. **[2]**

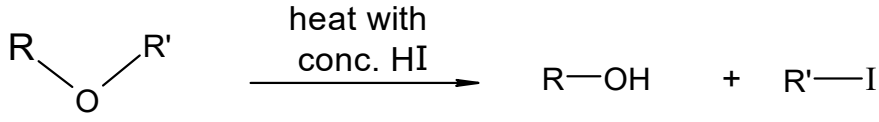
[TURN OVER

(c) Ethers have the general structure of $R_1–O–R_2$, where R_1 and R_2 are alkyl or aryl groups, for example, like methoxyethane.



methoxyethane

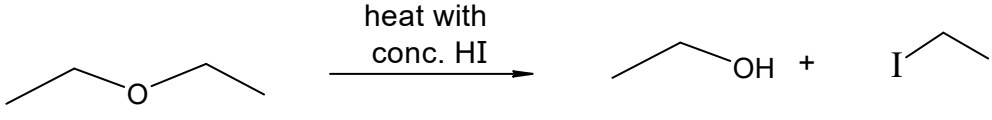
When reacted with hot concentrated solutions of hydroiodic acid, HI, ether can form an alkyl halide and an alcohol as shown in Fig. 3.2.



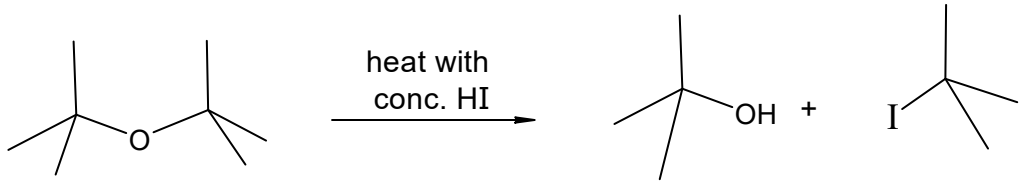
where R and R' are different alkyl groups

Fig. 3.2

Two examples are shown below.



diethyl ether



di-tert-butyl ether

(i) It was found that primary or secondary ethers, like diethyl ether, reacts via S_N2 mechanism while tertiary ethers, like di-tert-butyl ether, reacts via S_N1 mechanism.

Explain why this reaction proceeds mainly via:

1. S_N2 mechanism for primary or secondary ethers
2. S_N1 mechanism for tertiary ethers **[2]**

(ii) Compound **A**, $C_7H_{16}O$, is an ether, and it reacts with hot concentrated solution of hydroiodic acid, HI, to form compound **B**, $C_6H_{14}O$, and compound **C**.

B is found to rotate plane-polarised light. **B** forms a yellow precipitate with alkaline aqueous iodine. When **B** reacts with hot concentrated H_2SO_4, only compound **D**, C_6H_{12}, is formed.

C reacts with hot ethanolic $AgNO_3$ to form yellow precipitate.

With reference to Fig. 3.2, deduce a possible structure for **A**. Hence, suggest the structures for **B**, **C** and **D**. For each reaction, state the type of reaction described and explain what the information tells you about the functional groups present in each compound. **[8]**

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(d) Crown ethers are a class of cyclic ethers that have garnered significant attention due to their unique ability to form complexes with cations.

An example of a crown ether is 12-Crown-4 in Fig. 3.3.

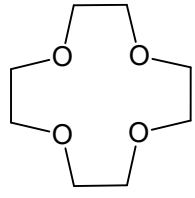

Fig. 3.3

12-Crown-4 forms complexes with various cations such as lithium ion, Li^+, in Fig. 3.4.

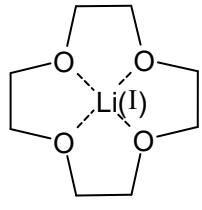

Fig. 3.4

(i) Suggest why 12-Crown-4 was found to be more selective in complexing with lithium ion, Li^+, than potassium ion, K^+. **[1]**

(ii) Suggest the type of interactions between 12-Crown-4 and Li^+. **[1]**

(iii) Crown ethers can be used to selectively remove caesium, a common component in nuclear waste, by complexing with caesium ions.

The radioactive decay of caesium-134 isotope is a first-order reaction with a half life of 2 years.

Use the following equations to calculate the percentage decrease of $^{134}Cs^+$ in a radioactive waste sample six months after it had been collected.

$$k = \frac{0.693}{t_{1/2}}$$

$$\frac{(^{134}Cs^+)_t}{(^{134}Cs^+)_o} = 10^{-\frac{kt}{2.3}}$$

where k is the rate constant,
$(^{134}Cs^+)_t$ is the amount of $^{134}Cs^+$ collected in the sample at time = t,
$(^{134}Cs^+)_0$ is the amount of $^{134}Cs^+$ collected in the sample at time = 0,
t is the time elapsed after the sample was collected. **[2]**

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